



Exploring strategies to generate Fitzpatrick skin type metadata for dermoscopic images using individual typology angle techniques

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Received: 3 February 2022 / Revised: 27 May 2022 / Accepted: 28 October 2022 /

Published online: 23 November 2022

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Abstract

When building machine learning models for dermatology one should strive to ensure that the resulting solutions do not exhibit any bias, which could cause a misdiagnosis of some patients – for example, due to the patient’s skin type being under-represented in the datasets used to build those models. Public dermatology datasets usually lack representativeness for certain skin tones, which are typically encoded using the Fitzpatrick skin type (a range of 1–6, where higher numbers mean darker skin tones). Moreover, those datasets usually lack metadata containing skin type information. To address the lack of such metadata, the Fitzpatrick skin type can be estimated based on pixel contents by computing the image’s Individual Typology Angle (ITA) and subsequently mapping ranges of ITA values to the corresponding Fitzpatrick skin type. This work focuses on computing the ITA for skin lesion images without binary masks indicating the location of the lesions. Four different approaches are proposed, experimentally evaluated using three datasets from the International Skin Imaging Collaboration (ISIC) and compared. The approaches were evaluated based on the estimated ITA value (essentially a *regression* task) and the resulting Fitzpatrick skin type (a *classification* task). Experimental results showed that the *Structured Patches* and *Center Cropped* methods performed best overall. The proposed algorithms can be used as a proxy for generating the Fitzpatrick skin type metadata for datasets that lack that metadata and without the need for segmentation masks.

Keywords Artificial intelligence · Machine learning · Dermatology · Fairness · Image analysis

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1 Introduction

Artificial Intelligence (AI) and Machine learning (ML) play a role in helping to improve the detection of malignant skin conditions in dermatology [3, 21, 30]. Researchers are likely to leverage public datasets when building these solutions. Several of these public databases have been found to lack diversity when it comes to darker skin types. Skin classification metadata is missing from several of these datasets. This information helps determine whether or not any AI/ML models have a bias for only certain skin types. Researchers must create fair models, since patient health might be jeopardized if they go untreated due to a model's bias. At the Sixth International Skin Imaging Collaboration (ISIC) Skin Image Analysis Workshop at CVPR 2021, only the PAD-UFES-20 [24] and Fitzpatrick 17k [13] datasets have skin type classification metadata. The following datasets from the same workshop lacked this metadata: Derm7pt [17], Dermofit Image Library [37], ISIC 2018 [6, 31], ISIC 2019 [5, 7, 31], ISIC 2020 [26], MED-NODE [12], PH2 [22], SD-128 [27], SD-198 [27], SD-260 [34].

The datasets that provided skin type metadata used a professional to evaluate each image. An alternative to this is to use the Individual Typology Angle (ITA) [33]. Using the ITA value of a whole image can be problematic in areas where skin lesions may be discolored compared to intact skin and alter the ITA value for this skin type. Some datasets provide skin lesion pixel annotation masks, which can be used to remove the skin lesion before computing the ITA value. Although not all datasets provide this pixel mask annotation, it can be challenging for a medical expert to develop accurate masks. Another approach could be to have a non-professional outline of the lesion, but some skin lesions require a dermatologist to fully understand the image [13].

In this paper, we provide new ways to estimate Fitzpatrick skin type metadata when both Fitzpatrick skin type metadata and skin lesion pixel mask data are lacking in the datasets. We appreciate that many researchers [10, 13, 19, 20, 28] have developed a AI/ML model to compute an ITA value or classify the patients Fitzpatrick's skin type, but we are trying to simplify our approach to intelligent ways of computing the ITA values for skin lesion images.

Figure 1 shows how to evaluate the ML classification models. A data set is divided into training, test, and evaluation datasets. An ML classification model is created and trained using the training dataset. The machine learning model is then evaluated using the testing and evaluation dataset. A confusion matrix and other metrics such as accuracy, precision, recall, F1 score, area under the curve, sensitivity, and specificity are used to evaluate the ML model performance. Each performance metric helps show how well the predicted class was chosen relative to the actual class.

Figure 2 represents the process of how our proposed approaches results are generated and evaluated. Although we use the same metrics as the classical machine learning model in Fig. 1, we do not use machine learning for our predictions. The ITA value is calculated from an image using (1) and is used together with our methods in Section 4. In the upper part of Fig. 2, the results are treated as a regression problem, where the calculated ITA of an image is compared to the 'baseline' for all proposed methods. The 'baseline' is computed because the datasets do not provide ITA metadata. The 'baseline' ITA is calculated from an image where the skin lesion is removed. RSME and MAE are used as metrics for comparing the calculated ITA (for each approach) and the 'baseline' ITA. The lower part of a Fig. 2, the results are treated a classification problem. The ITA value calculated from an image and the 'baseline' ITA are mapped to the Fitzpatrick skin type using the conversion Table 2.

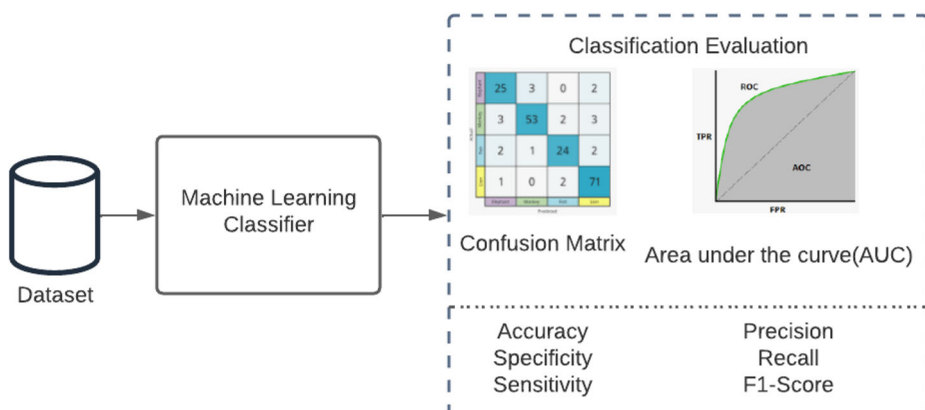


Fig. 1 A typical process of evaluating Machine Learning models

These Fitzpatrick skin type results are then compared using the same evaluation metrics as a classical machine learning model.

In summary, the key contributions of this paper are as follows:

- *Novel approaches to compute ITA values in the absence of segmentation masks.* Four different methods for computing the ITA value of a skin lesion image without segmentation mask are proposed and evaluated.
- *Systematic evaluation of the impact of skin lesion size on the quality of the results.* For any of the proposed methods, the size of the skin lesion is a contributing negative factor when trying to estimate the Fitzpatrick skin type.
- *Open source library¹:* A publicly available Python library that calculates the ITA value of an image and can map this ITA value to the Fitzpatrick skin type from multiple available conversion tables has been created to encourage reproducible research.

2 Related work

The field of skin lesion image analysis has benefited from the availability of public annotated datasets [15, 25, 35, 36] as well as associated challenges² [2, 5, 6, 14, 31]. The public availability of relatively large datasets has led to advancements in skin lesion detection, skin lesion segmentation, skin cancer classification, skin tone fairness evaluation and many other relevant research topics under the scope of AI in dermatology.

These datasets and challenges, however important, have their shortcomings. Particularly relevant to this paper is the problem of representative samples across all Fitzpatrick skin types. It's a known problem that many of the public datasets have significantly more lighter skin images than darker skin images [8, 9, 18]. This becomes problematic when using these datasets to build AI/ML models which could have a bias towards the lighter skin. Kamulegeya et al. [16] needed to create their own dataset of darker skin types to build

¹<https://pypi.org/project/derm-ita/>

²Examples of challenges include <https://grand-challenge.org/challenges/> and <https://www.kaggle.com/>

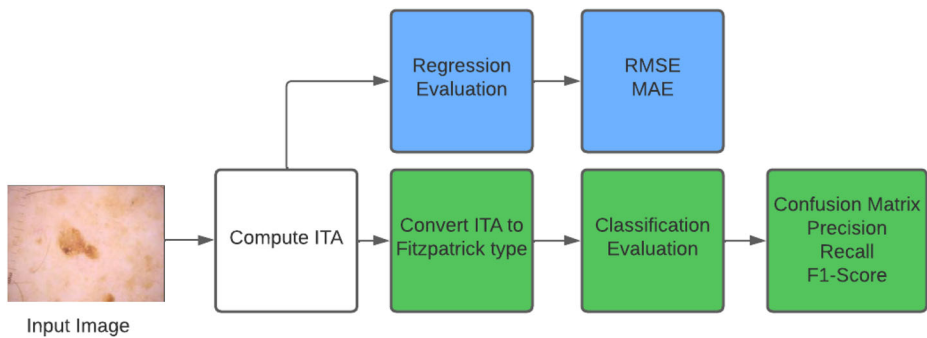


Fig. 2 The process of taking an image and computing the ITA value. In the blue boxes the ITA value is evaluated against the ‘baseline’ ITA value. In the green boxes the ITA value is converted to the Fitzpatrick type and then evaluated against the ‘baseline’ Fitzpatrick type

an AI/ML model for Uganda. Building applications for areas around the world where the majority of the population has darker skin will take more work to be as successful.

Ideally, skin image datasets should have metadata reporting, among other things, the Fitzpatrick type of each image. Absent that, if at least ground truth information for lesion segmentation (typically in the form of binary masks covering the lesion region) are provided, the skin type can then be inferred from pixel values that fall outside the lesion area using techniques such as the computation of ITA [32, 33]. This is a relatively recent research effort that has been explored by some research groups, as summarized next.

Kinyanjui et al. [19, 20] used Mask R-CNN to segment the skin lesion, computed the ITA value and then calculated the Fitzpatrick skin tone. They also expanded the standard 6 class scale found on Table 1 and expanded to a total 8 types because determined that types 2,3, and 4 had the largest distribution of people in the ISIC and SD-198 datasets and felt the need to further refine this set. With this data they evaluated and determined that both datasets under-represent the darker skin tones.

Tadesse et al. [28] created an automated pipeline for evaluating images used in dermatology datasets for the skin tones. They also used Mask R-CNN to segment the lesion and then computed the ITA value and then mapped to the Fitzpatrick skin types. Their evaluation showed that images only 10.7% had darker skin tones proving a lack of representation in dermatology material.

Groh et al. [13] worked with dermatologist to help curate the Fitzpatrick17k and annotate all the imaged with their respected Fitzpatrick skin types. These images include more than use dermoscopic images which are only in the ISIC datasets. This team took a different

Table 1 Original mapping of ITA ranges to the six class Fitzpatrick skin types [4]

ITA Range	Skin type Category
$55^\circ < \text{ITA}$	Type 1
$41^\circ < \text{ITA} \leq 55^\circ$	Type 2
$28^\circ < \text{ITA} \leq 41^\circ$	Type 3
$19^\circ < \text{ITA} \leq 28^\circ$	Type 4
$10^\circ < \text{ITA} \leq 19^\circ$	Type 5
$\text{ITA} \leq 10^\circ$	Type 6

approach and created a YCbCr masked to focus on the skin instead of using a Mask R-CNN or another segmentation method to remove the skin lesions. Once the image has been masked, the ITA value is computed and compared to the ITA value computed on a full image. Their approach showed promising results over the full image. They also made their own range values for the Fitzpatrick skin types shown on Table 2 to increase their accuracy compared to the full image results.

Sauman Das [10] only evaluated the Fitzpatrick types 1, 2, and 3 and dense hair classifier. They found using the ISIC's HAM10000 dataset it was a challenge distinguishing between types 4, and 5 compared to lighter skins types with dark and dense hair. They also developed a way to label the Fitzpatrick skin types and out performed [13] in the Fitzpatrick17k dataset from 53.3% to 60.61% accuracy. They used a method of blurring to segment out the skin lesion and then estimate the Fitzpatrick skin type.

The work proposed in this paper adapts and extends the work of Groh et al. [13]. In the absence of masks our proposed methods will strategically compute the ITA value of an image using four different methods. The ITA value will then be mapped to the Fitzpatrick skin type using Table 2. It will be described in detail in the next section.

3 Proposed work

3.1 Methodology

The six Fitzpatrick skin types were created to categorically track the results of different skin types and how they react to sun exposure. [11]. Over 30 years ago, Chardon, Cretois, and Hoursea [4] developed the ITA calculation and used this result to map to each of the six Fitzpatrick skin types. Each skin type, as seen in Table 1, maps to a range of ITA values.

In 2021, Groh et al. [13] updated the ranges using the following method. They mask each image using a special YCbCr mask and then compute the ITA value for each image on the Fitzpatrick 17k dataset and then map to the Fitzpatrick skin type. They started with Table 1 but then adjusted the ranges to reduce the error comparing to the Fitzpatrick meta-data provided in the Fitzpatrick 17k dataset, producing the mappings shown in Table 2. This is relevant because in our work we also want to classify images from ISIC datasets into Fitzpatrick skin types. From this point on, we refer to the mapping and work done by Groh et al. [13] as starting points for the methods proposed in this paper.

Both mapping ranges can be visually compared in Fig. 3. In both charts it can be seen that Type 1 and Type 6 have quite large ranges compared to the others. Types 2, 3, 4 and 5 have small ranges which make it problematic to correctly classify the Fitzpatrick type. This is especially important if something on the image throws off the ITA calculation as described

Table 2 Alternative mapping of ITA ranges to the six class Fitzpatrick skin types [13]

ITA Range	Skin type Category
$40^\circ < \text{ITA}$	Type 1
$23^\circ < \text{ITA} \leq 40^\circ$	Type 2
$12^\circ < \text{ITA} \leq 23^\circ$	Type 3
$0^\circ < \text{ITA} \leq 12^\circ$	Type 4
$-25^\circ < \text{ITA} \leq 0^\circ$	Type 5
$\text{ITA} \leq -25^\circ$	Type 6

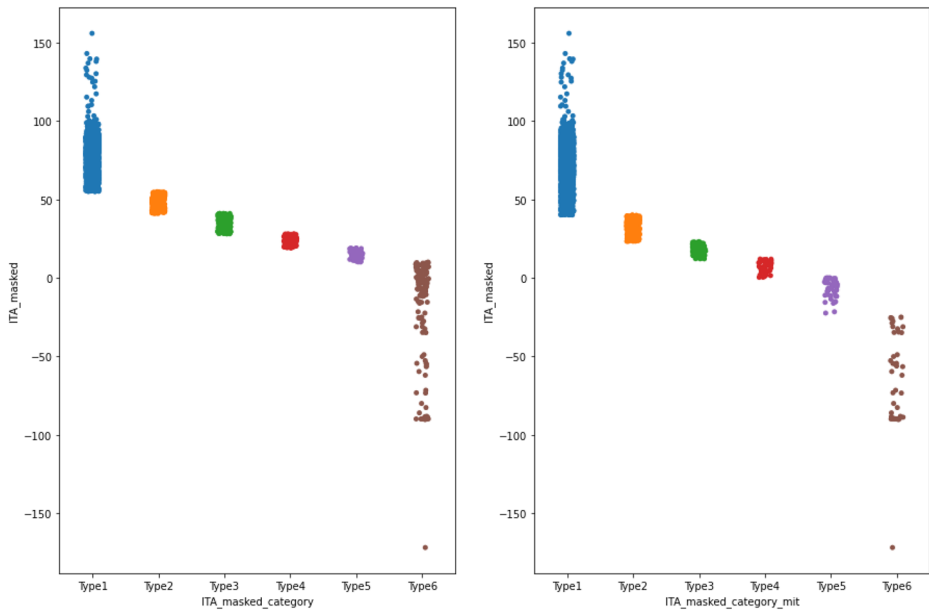


Fig. 3 Strip plot of ITA values, computed from a whole image, for each of the skin types using conversion Tables 1 and 2 from [13] on the ISIC 2017 dataset

in the limitations part in Section 7. Lastly, it should be noted that Type 1 (lighter skin tones) has a significantly larger population compared to the other types, which is a well-known problem that has fairness implications beyond the scope of this paper [8, 9, 18].

The ITA computation relies upon the color type CIELab because it uses the L and b values. The L represents the luminosity (L) and the b value to represent the yellowness of the image shown in Fig. 4. The ITA value is calculated using (1) [23]. A pre-processing step, not shown in the equation, omits the black pixels representing the skin lesion mask from the formal ITA equation. We assume that the skin lesion, represented by this mask, would be a different color and is worth removing.

$$ITA = \arctan\left(\frac{L^* - 50}{b^*}\right) \times \frac{180^\circ}{\pi} \quad (1)$$

After the ITA value is calculated, Table 2 is used to map the Fitzpatrick skin type.

3.2 Challenges

In this work the ITA value is used as a proxy for skin type classification. When computing the ITA value, we need to consider that in most cases, the skin lesion is different from the patient's skin color. This is problematic because the different skin lesion colors could skew the ITA value that is trying to represent the skin type and place someone in a different Fitzpatrick skin type. Some datasets provide skin lesion mask annotation, which can be used to remove the skin lesion and solve our skin lesion color issue. In this work, we assume that such masks are not available, which brings up the following challenges:

1. **Image and lesion size:** When a skin lesion grows large enough to take up most of the image, it can be challenging to capture the skin accurately.

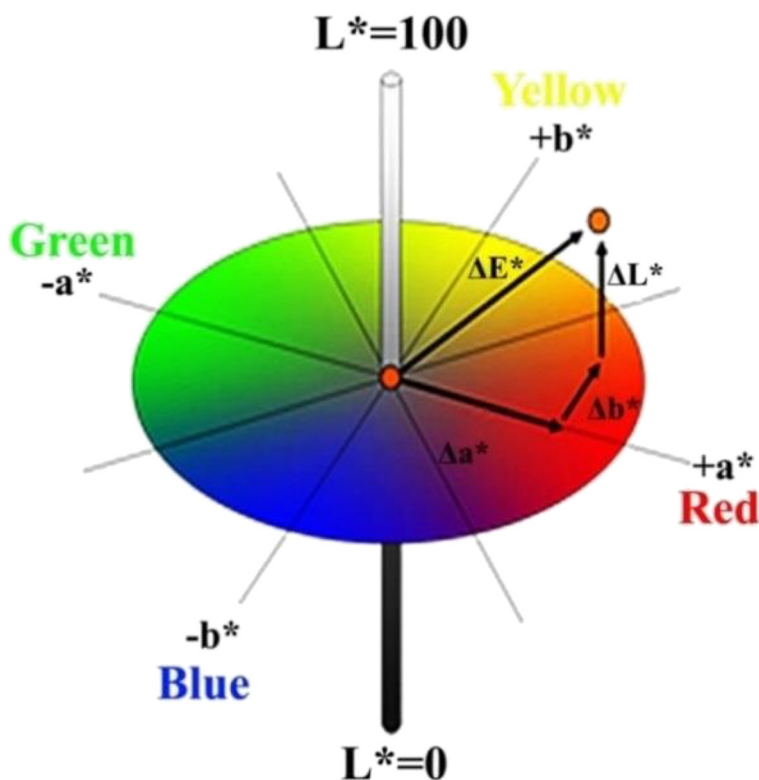


Fig. 4 Visual representation showing the components of the CIE Lab color space [29]. The important values used in ITA equation at L and positive b values. L represents the luminosity or brightness and the positive b values represent the yellowness of the image

2. Dark corners: When capturing dermoscopic images, the corners can be significantly darker due to how the image is taken.
3. Stickers and other artifacts: Some images might have stickers with different colors or pen markings outlining the image's main subject. These situations can cause issues when computing the ITA value as they are all of a different color than the skin color, distorting the ITA value.
4. Finally, because of the convenience of being remote from a medical professional, teledermatology has grown in popularity. In these cases, patients are required to take pictures of their skin lesions. The lighting or the quality of the camera could result in dim, bright, blurry images. The results could skew the ITA values, misinterpreting the correct skin type. Different approaches have been studied to limit these issues [1]. A person could self-adjust to these conditions, but it could be challenging for a computer to determine.

3.3 Four methods for computing ITA values without lesion segmentation masks

This work proposes four methods for computing an image's ITA value and Fitzpatrick skin type label when a dataset lacks the Fitzpatrick skin type metadata and the pixel mask annotation:

- Full image
- Center cropped
- Structured patches
- Random patches

The *Full Image* approach computes the ITA value on the whole image and does not break the image down into different patches. The full image is a simple approach that takes minimal effort to set up but the skin lesion can skew the ITA result because the lesion is likely a different color than the skin. Example images can be viewed in Fig. 5 in column (a).

In the cases of *Center Cropped*, *Structured Patches*, and *Random Patches*, a *patch* is referred to as a predetermined $N \times N$ pixels subimage. The ITA value is computed for each patch. Each of these methods will have a set of patches and the median of the set of patches is selected as the ITA value of that image.

The *Center Cropped* approach ignores the center portion of the image, assuming that the skin lesion would exist in the center, thus reducing any bias the lesion color might have. A few equations were created on the basis of height and width to determine what to capture or ignore. Using 35% of the width and height will obtain an offset value in the (2) and (3). Then the midpoint was computed by dividing the width and height by 2 in (4) and (5). Then the offset is used to create the condition when to ignore these sets of patches in (6) and (7). When patches fall outside these conditions, the ITA will be computed for those patches. The median ITA value is then selected to represent the ITA value of the image. The result can be viewed in Fig. 5 in column (c).

$$w_o = w \times 0.35 \quad (2)$$

$$h_o = h \times 0.35 \quad (3)$$

$$mid_w = w/2 \quad (4)$$

$$mid_h = h/2 \quad (5)$$

$$mid_w - w_o \leq w < mid_w + w_o \quad (6)$$

$$mid_h - h_o \leq h < mid_h + h_o \quad (7)$$

The *Structured Patches* approach takes patches around the border of the image. This border is 1 patch thickness around all four sides. The center of the image was not sampled as we assume that the skin lesion is in that portion of the image. The result can be viewed in Fig. 5 in column (d).

The *Random Patches* approach will randomly sample patches all over the image. This approach should include some skin sampling but will likely also sample the lesion depending on the lesion size. In early testing the results did not vary much when randomly sampling between 10–75% of the images. Based on these early experiments 25% of the image was selected. The result can be viewed in Fig. 5 in column (e).

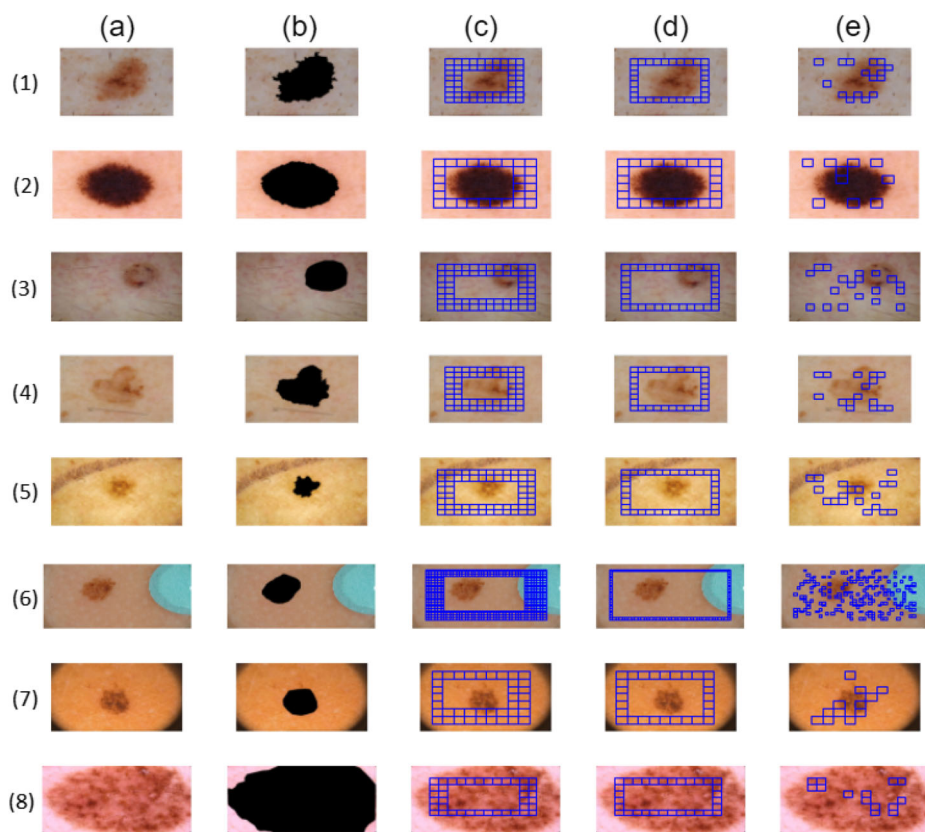


Fig. 5 Sample of all the different approaches on the ISIC 2016 dataset. Column (a) is whole image, (b) is masked, (c) is center cropped, (d) is structured patches, (e) is random patches. In columns (c), (d), (e), each blue square is considered a patch. In this sample, there are artifacts such as stickers and large skin lesions that can negatively affect the computation of these approaches

4 Methods

4.1 Overview

Figure 6 shows an overview of the pipeline process used in this study. Before each approach, each image boarder was trimmed by 4% pixels to avoid any black corners that can typically appear in dermoscopic images (e.g., Fig. 5 row 7). Increasing the border percentage to around 8% has negatively impacted the results because the lesion becomes closer to the edge, and there is less skin to sample. Anything less than 4% tended to include the dark corners, negatively impacting the ITA value. Based on those preliminary results, all datasets had 4% removed from each image before the ITA calculation phase.

4.2 Patches and patch sizes

Patch sizes for each dataset were hand selected based on performance (see Table 3). The 2017 ISIC dataset was already preprocessed [35] where the images were resized to

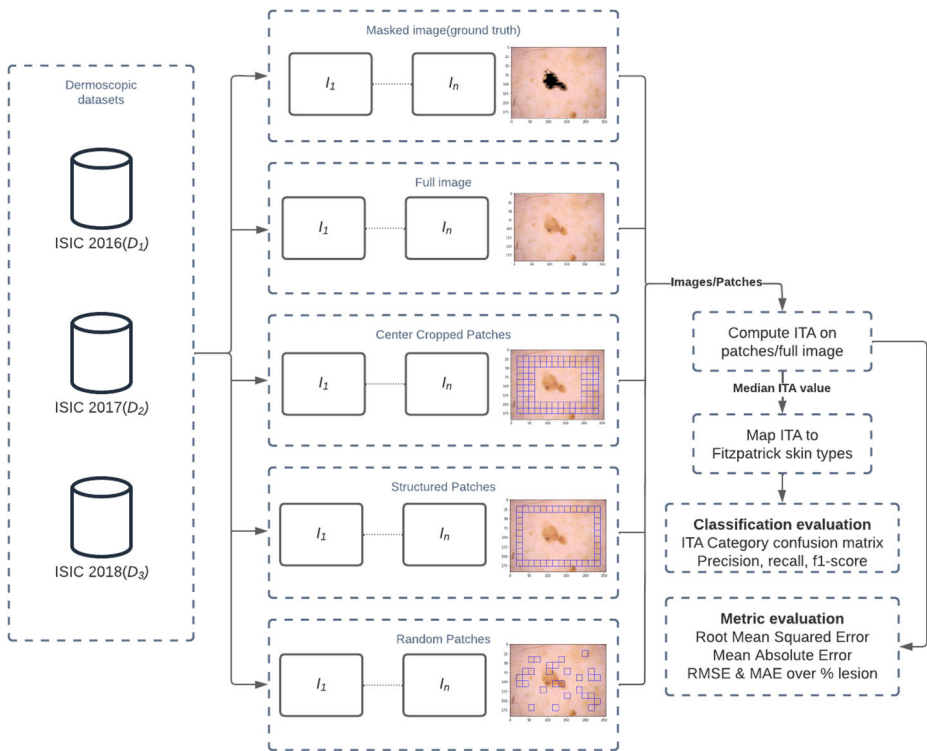


Fig. 6 Overview of the processing and evaluating of the different approaches. Images are processed for each of the proposed approach to compute the ITA value. The ITA value is then compared against the baseline ITA. The ITA value is also mapped the the Fitzpatrick skin type and compared against the baseline Fitzpatrick skin type

192×256, and a patch size 8×8 was selected. The 2018 ISIC dataset had very detailed images of up to 4,499 × 6,748 pixels, and a patch size of 32×32 was selected. Increasing the patch size on the datasets with larger images decreased the ITA values' runtime. The 2016 and 2018 datasets image sizes were not all the same, which is depicted in Fig. 5. When comparing column (d) row 1 and column (d) row 6, the patch outlines appear to be different sizes, but in reality, they are all 8×8.

Once all the of pre-processing are complete, every dataset image are processed using all four different approaches to compute the ITA value. When an approach used patches, the

Table 3 Dataset details on number of images, their sizes and selected patch size

Dataset	Image count	Image Sizes (height,width)			Patch size
		Min	Max	Average	
ISIC 2016	1277	(566,679)	(2848,4288)	(1114.45,1530.02)	8×8
ISIC 2017	2750	(192,256)	(192,256)	(192,256)	8×8
ISIC 2018	2592	(566,679)	(4499,6748)	(2166.79,3188.26)	32×32

median value of the set of ITA values was selected. The ITA value is mapped to conversion Table 2 to estimate the skin type.

4.3 Datasets

Since most dermoscopic datasets do not provide the Fitzpatrick skin type metadata, our criteria for selecting a dataset was that it should have pixel mask annotation for removing the skin lesion from the image. For our testing, we selected datasets that provided a pixel mask annotation for skin lesions (ISIC 2016, ISIC 2017, and ISIC 2018). Typically, these masks make analyzing the lesion easier by creating an outline of the skin lesion to allow easy removal of the skin. Since the skin is the subject of our testing, the mask was inverted to generate an image that will have the skin lesion cut out. The result can be viewed in Fig. 5 in column (b), where the lesion is completely removed, and only the skin remains. Since the ISIC datasets lack Fitzpatrick skin type metadata for our testing, we consider these images with the skin lesion removed as our baseline. When computing the ITA values on the baseline images, the black pixels of 0 RGB values were ignored. Also, in all the approaches, the border is removed from being sampled because a high percentage of dermoscopic images have dark corners. Removing these borders reduces any disparities in the ITA computation.

We chose the ISIC challenge datasets from 2016 [14], 2017 [2, 5], and 2018 [6, 31]. These datasets were chosen because they included pixel mask annotations of the skin lesion. That meant we did not need to create a solution to remove the skin lesion from the image for our computations. Because these datasets do not provide the Fitzpatrick skin type, we use the image with the skin lesion removed as a baseline.

4.4 Language and computational environment

The computation of the ITA value has been packaged into a Python library³ and the source has been made publicly available⁴. For the purposes of these experiments, everything was run on Google Colaboratory⁵ with no extra special back-end hardware.

4.5 Computational complexity analysis

Each approach was tracked how long it would take to compute the ITA value and estimate the Fitzpatrick skin type shown in Table 4. In all datasets, the Center Cropped approach had the fastest time. ISIC 2018 dataset had the longest run times because the average image size was significantly larger, meaning more patches and more ITA computations per image relative to the ISIC 2016 and ISIC 2017 datasets. Both *Center Cropped* and *Structure Patches* generally had faster run times because they sampled fewer of the overall pixels of the image. The other approaches either had more complexity, such as removing the skin lesion with the mask or sampled more significant portions of the image in both *Full image* and *Random Patches*. Overall, each approach's runtime was relatively similar and would not be a significant factor when selecting an approach.

³<https://pypi.org/project/derm-ita/>

⁴https://github.com/AdamCorbinFAUPhD/derm_ita

⁵<https://colab.research.google.com/>

Table 4 Computational run times on datasets across all approaches

Dataset	Approach	Runtime
ISIC 2016	Full image	19min 14s
	Center Cropped	16min 34s
	Masked Image	22min 5s
	Structured Patches	21min 29s
	Random Patches	24min 36s
ISIC 2017	Full image	46.1s
	Center Cropped	39.8s
	Masked Image	3min 1s
	Structured Patches	3min 2s
	Random Patches	3min 27s
ISIC 2018	Full image	2h 42min 32s
	Center Cropped	2h 17min 32s
	Masked Image	2h 34min 8s
	Structured Patches	2h 25min 55s
	Random Patches	2h 47min 38s

4.6 Figures of merit

As described earlier in this paper (Fig. 2), we performed two types of quantitative evaluation:

1. Regression: where the goal is to assess how well each method performs in predicting the (continuous, real) value of the ITA of the image. Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE) are used to compare the ITA values between the baseline (masked image) and each of the four proposed approaches.
2. Classification: where the goal is to evaluate each method after the (continuous, real) ITA value has been mapped to the corresponding Fitzpatrick type using Table 2. A confusion matrix is created for comparing the performance between the baseline and each of the four proposed approaches. For each case, the following metrics were computed: precision, recall, and f1-score.

5 Results

Three different evaluations were performed to assess the feasibility of the proposed approaches in the absence of labeled (i.e., masked) images indicating which pixels belong to the lesion and which ones do not.

5.1 ITA estimation using regression figures of merit

The first set of results treats the problem as a *regression* problem, namely the estimation of an actual number (the ITA for each image, as established by computing the ITA using (1) on pixels outside of the mask). It reflects how well each approach performed by comparing the computed ITA value relative to the ITA value of the masked image ground truth. In the evaluation, popular literature measures such as the Root Mean Squared Error (RMSE) and

Table 5 Metric results using Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE) across all dataset tests

Dataset	Approach	RMSE	MAE
ISIC 2016	Full image	28.298	15.307
	Center Cropped	20.581	9.505
	Structured Patches	25.546	11.702
	Random Patches	28.420	14.882
ISIC 2017	Full image	29.193	13.008
	Center Cropped	20.640	8.195
	Structured Patches	20.043	8.133
	Random Patches	28.541	12.848
ISIC 2018	Full image	29.181	14.388
	Center Cropped	22.044	9.252
	Structured Patches	21.333	9.361
	Random Patches	29.435	14.181

the Mean Absolute Error (MAE) were used. Table 5 shows the results for each combination of method and dataset, where the best results are highlighted in **bold**.

5.2 The impact of lesion size on ITA estimation

The second set of results takes the first evaluation and groups the results based on the size of the lesion (relative to the image). Figure 7 shows results of the RMSE and MAE evaluation

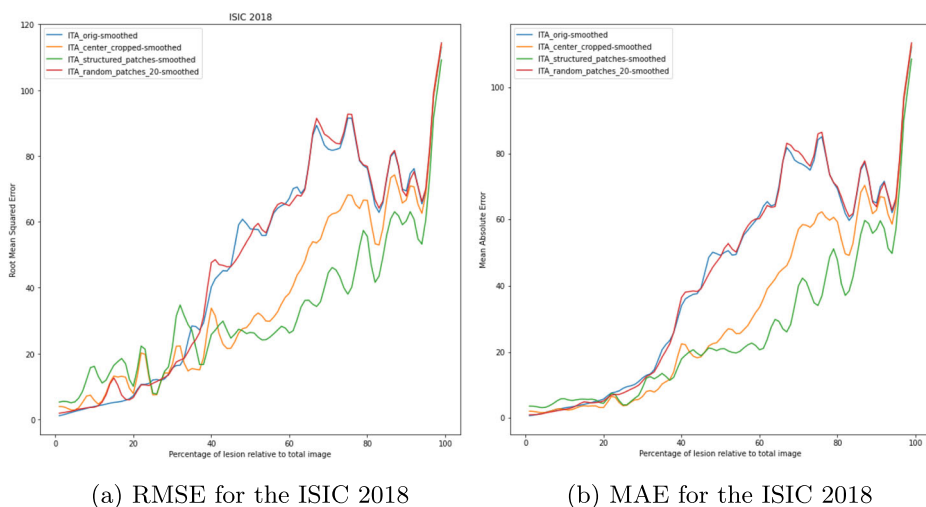
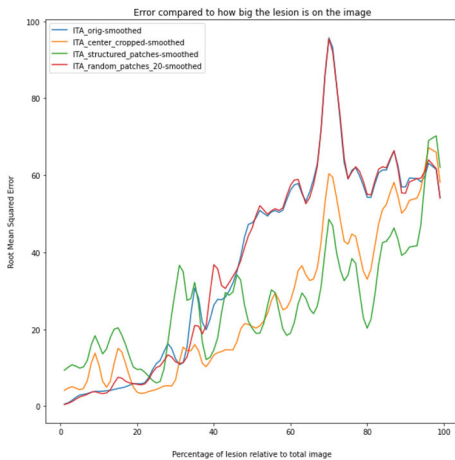
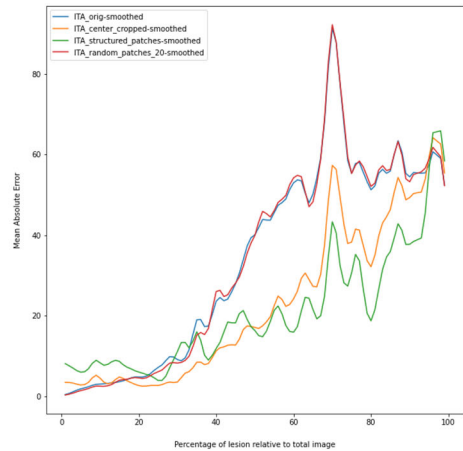


Fig. 7 Selected examples of how MAE and RMSE vary with respect to the relative size of the lesion in images from two different datasets



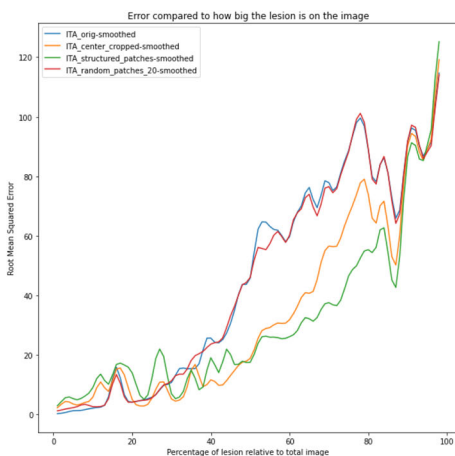
(a) RMSE for the ISIC 2016



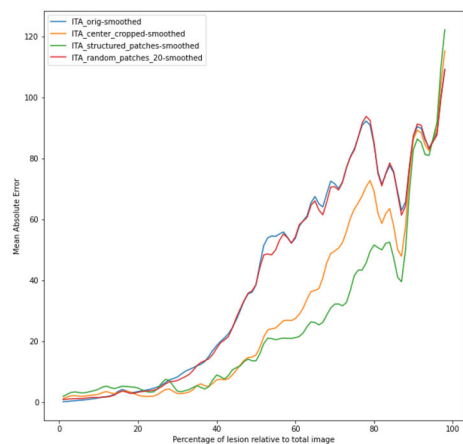
(b) MAE for the ISIC 2016

Fig. 8 Selected examples of how MAE and RMSE vary with respect to the relative size of the lesion in images from two different datasets

metrics measured against the skin lesion size for the ISIC 2018 dataset. Similar results viewed for ISIC 2016 in Fig. 8 and for ISIC 2017 in Fig. 9. The fluctuations of these curves can be caused by the increasing size of the skin lesion along with the placement of the skin lesion. Further discussion can be found on in Section 6.



(a) RMSE for the ISIC 2017



(b) MAE for the ISIC 2017

Fig. 9 Selected examples of how MAE and RMSE vary with respect to the relative size of the lesion in images from two different datasets

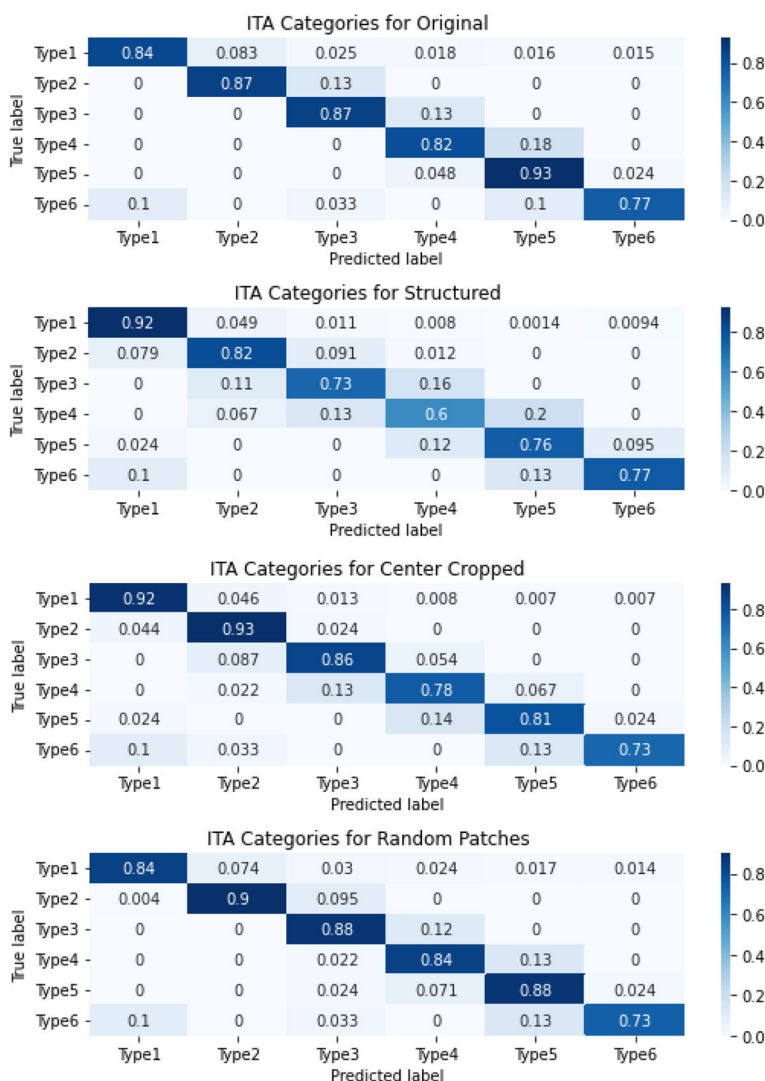


Fig. 10 The Normalized Confusion matrix compares the predicted ITA skin types across all approaches for the ISIC 2018 dataset. The Table 2 was used to create the Fitzpatrick labels

5.3 Fitzpatrick skin type estimation using classification figures of merit

The third set of results treats the problem as a *classification* problem. Once the ITA value has been computed, it can then be mapped to the Fitzpatrick skin types in Table 2. The classification performance of all of the approaches used in this study for the ISIC 2018 dataset is illustrated in Fig. 10 which compare the masked image baseline. The same comparison for ISIC 2016 is represented on Fig. 11 and for ISIC 2017 in Fig. 12. For more details, cat-

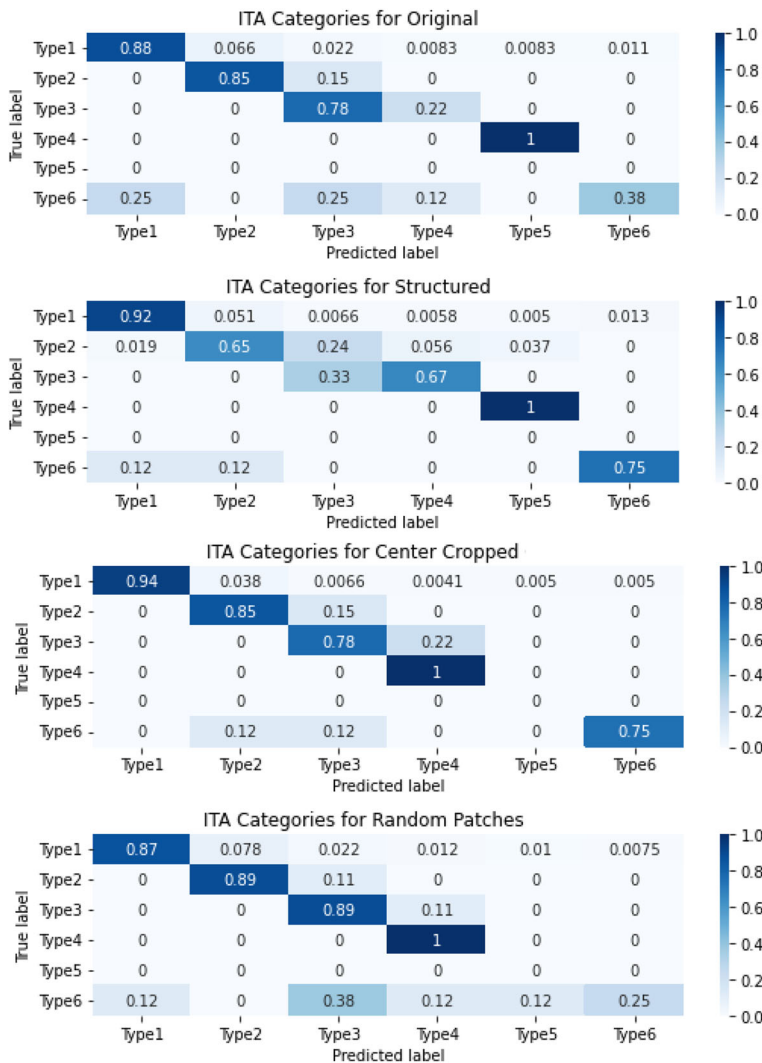


Fig. 11 The Normalized Confusion matrix compares the predicted ITA skin types across all approaches for the ISIC 2016 dataset. The Table 2 was used to create the Fitzpatrick labels

egorization metrics for the ISIC 2016 dataset can be found on Tables 6, 7, 8 and 9 for ISIC 2017 on Tables 10, 11, 12 and 13 and for ISIC 2018 on Tables 14, 15, 16 and 17.

6 Discussion

From the results in Section 5 we will break up the discussion between the three evaluations.

- The *regression* evaluation of computing the ITA value showed both *Center Cropped* and *Structured Patches* were the top performers in any dataset. In both of these approaches, the objective was to avoid capturing the skin lesion when computing the ITA value,

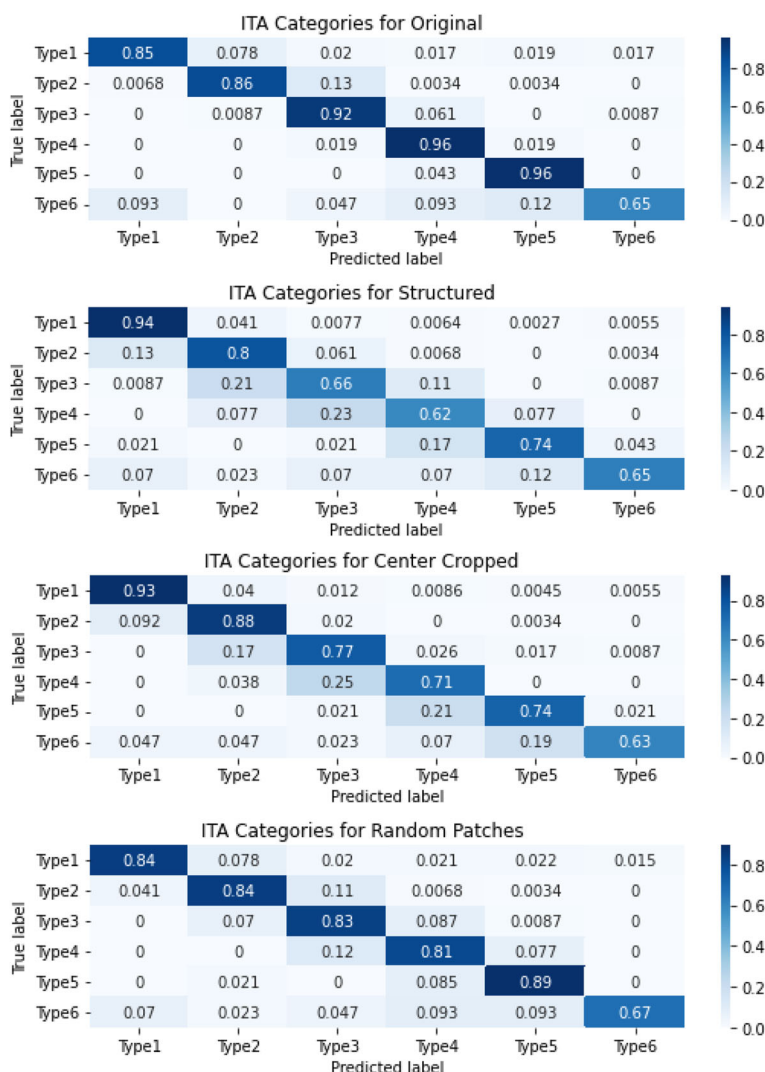


Fig. 12 The Normalized Confusion matrix compares the predicted ITA skin types across all approaches for the ISIC 2017 dataset. The Table 2 was used to create the Fitzpatrick labels

which helped their performance over the *Full image* and *Random Patches*. These lower-performing approaches included the skin lesion as part of the ITA computation, which degraded their performance relative to the ground truth. For ISIC 2016, the *Center Cropped* approach was the clear winner, which might be a result of the dataset having skin lesions centered and smaller. The difference in values between the best and second best methods was not as significant for the other two datasets. These other two datasets were curated to evaluate more complex problems, resulting in similar results. These complex problems involve the evaluation of different types of skin cancer and segmentation of skin lesions. The 2016 ISIC dataset had a simple task of determining whether a skin lesion was melanoma or benign.

Table 6 Metric results using ISIC 2016 with the approach whole image and using the Table 2 for the classification estimation

ISIC 2016 Full image				
	Precision	Recall	f1-score	Support
Type 1	1.00	0.88	0.94	1205
Type 2	0.37	0.85	0.51	54
Type 3	0.16	0.78	0.27	9
Type 4	0.00	0.00	0.00	1
Type 5	0.00	0.00	0.00	0
6	0.19	0.38	0.25	8
accuracy			0.88	1277
macro avg	0.29	0.48	0.33	1277
weighted avg	0.96	0.88	0.91	1277

Table 7 Metric results using ISIC 2016 with the Center Cropped approach and using the Table 2 for the classification estimation

ISIC 2016 Center cropped				
	Precision	Recall	f1-score	Support
Type 1	1.00	0.94	0.97	1205
Type 2	0.49	0.85	0.63	54
Type 3	0.29	0.78	0.42	9
Type 4	0.12	1.00	0.22	1
Type 5	0.00	0.00	0.00	0
Type 6	0.50	0.75	0.60	8
Accuracy			0.94	1277
Macro avg	0.40	0.72	0.47	1277
Weighted avg	0.97	0.94	0.95	1277

Table 8 Metric results using ISIC 2016 with the structured patches approach and using the Table 2 for the classification estimation

ISIC 2016 Structured patches				
	Precision	Recall	f1-score	Support
Type 1	1.00	0.92	0.96	1205
Type 2	0.36	0.65	0.46	54
Type 3	0.12	0.33	0.18	9
Type 4	0.00	0.00	0.00	1
Type 5	0.00	0.00	0.00	0
Type 6	0.27	0.75	0.40	8
Accuracy			0.90	1277
Macro avg	0.29	0.44	0.33	1277
Weighted avg	0.96	0.90	0.93	1277

Table 9 Metric results using ISIC 2016 with the random patches approach and using the Table 2 for the classification estimation

ISIC 2016 Random patches				
	Precision	Recall	f1-score	Support
Type 1	1.00	0.87	0.93	1205
Type 2	0.34	0.87	0.49	54
Type 3	0.20	0.89	0.32	9
Type 4	0.05	1.00	0.09	1
Type 5	0.00	0.00	0.00	0
Type 6	0.18	0.25	0.21	8
Accuracy			0.87	1277
Macro avg	0.29	0.65	0.34	1277
Weighted avg	0.96	0.87	0.90	1277

Table 10 Metric results using ISIC 2017 with the Full Image approach and using the Table 2 for the classification estimation

ISIC 2017 Full image				
	Precision	Recall	f1-score	Support
Type 1	1.00	0.85	0.92	2200
Type 2	0.59	0.86	0.70	293
Type 3	0.56	0.92	0.70	115
Type 4	0.49	0.96	0.65	52
Type 5	0.48	0.96	0.64	47
Type 6	0.42	0.65	0.51	43
Accuracy			0.85	2750
Macro avg	0.59	0.87	0.69	2750
Weighted avg	0.91	0.85	0.87	2750

Table 11 Metric results using ISIC 2017 with the Center cropped approach and using the Table 2 for the classification estimation

ISIC 2017 Center cropped				
	Precision	Recall	f1-score	Support
Type 1	0.99	0.93	0.96	2200
Type 2	0.70	0.88	0.78	293
Type 3	0.65	0.77	0.71	115
Type 4	0.51	0.71	0.60	52
Type 5	0.62	0.74	0.68	47
Type 6	0.66	0.63	0.64	43
Accuracy			0.91	2750
Macro avg	0.69	0.78	0.73	2750
Weighted avg	0.92	0.91	0.91	2750

Table 12 Metric results using ISIC 2017 with the Structured patches approach and using the Table 2 for the classification estimation

ISIC 2017 Structured patches				
	Precision	Recall	f1-score	Support
Type 1	0.98	0.94	0.96	2200
Type 2	0.66	0.80	0.73	293
Type 3	0.60	0.66	0.63	115
Type 4	0.44	0.62	0.52	52
Type 5	0.70	0.74	0.72	47
Type 6	0.64	0.65	0.64	43
Accuracy			0.90	2750
Macro avg	0.67	0.74	0.70	2750

Table 13 Metric results using ISIC 2017 with the Random Patches approach and using the Table 2 for the classification estimation

ISIC 2017 Random patches				
	Precision	Recall	f1-score	Support
Type 1	0.99	0.84	0.91	2200
Type 2	0.58	0.84	0.69	293
Type 3	0.53	0.83	0.65	115
Type 4	0.39	0.81	0.52	52
Type 5	0.42	0.89	0.57	47
Type 6	0.46	0.67	0.55	43
Accuracy			0.84	2750
Macro avg	0.56	0.82	0.65	2750
Weighted avg	0.90	0.84	0.86	2750

Table 14 Metric results using ISIC 2018 with the Full Image approach and using the Table 2 for the classification estimation

ISIC 2018 Full image				
	Precision	Recall	f1-score	Support
Type 1	1.00	0.84	0.91	2131
Type 2	0.55	0.87	0.68	252
Type 3	0.48	0.87	0.62	92
Type 4	0.41	0.82	0.55	45
Type 5	0.46	0.93	0.61	42
Type 6	0.41	0.77	0.53	30
Accuracy			0.85	2592
Macro avg	0.55	0.85	0.65	2592
Weighted avg	0.91	0.85	0.86	2592

Table 15 Metric results using ISIC 2018 with the Center cropped approach and using the Table 2 for the classification estimation

ISIC 2018 Center cropped				
	Precision	Recall	f1-score	Support
Type 1	0.99	0.92	0.95	2131
Type 2	0.69	0.93	0.79	252
Type 3	0.67	0.86	0.75	92
Type 4	0.56	0.78	0.65	45
Type 5	0.61	0.81	0.69	42
Type 6	0.58	0.73	0.65	30
Accuracy			0.91	2592
Macro avg	0.68	0.84	0.75	2592
Weighted avg	0.93	0.91	0.92	2592

Table 16 Metric results using ISIC 2018 with the Structured patches approach and using the Table 2 for the classification estimation

ISIC 2018 Structured patches				
	Precision	Recall	f1-score	Support
Type 1	0.99	0.92	0.95	2131
Type 2	0.64	0.82	0.72	252
Type 3	0.56	0.73	0.63	92
Type 4	0.40	0.60	0.48	45
Type 5	0.67	0.76	0.71	42
Type 6	0.49	0.77	0.60	30
Accuracy			0.89	2592
Macro avg	0.62	0.77	0.68	2592
Weighted avg	0.92	0.89	0.90	2592

Table 17 Metric results using ISIC 2018 with the Random patches approach and using the Table 2 for the classification estimation

ISIC 2018 Random patches				
	Precision	Recall	f1-score	Support
Type 1	1.00	0.84	0.91	2131
Type 2	0.59	0.90	0.71	252
Type 3	0.47	0.88	0.62	92
Type 4	0.37	0.84	0.51	45
Type 5	0.45	0.88	0.59	42
Type 6	0.42	0.73	0.54	30
Accuracy			0.85	2592
Macro avg	0.55	0.85	0.65	2592
Weighted avg	0.91	0.85	0.87	2592

- Figure 7 evaluates the performance using RMSE and MAE where the images were grouped based on the size of the skin lesion from the ISIC 2018 dataset. Most of the approaches are performed equally when the lesion is small. When the lesion becomes larger, around 40–85% of the image, the *Structured Patches* and *Center Cropped* outperform the *Full image* and *random approaches*. Once the lesion starts to reach 85% most, all approaches degrade performance. This test shows that it is challenging to capture the portion that is only skin. At the higher skin lesion percentages, the *Structured Patches* and *Center Cropped* approaches would look primarily at the lesion instead of the desired skin, which significantly reduced their performance. We also notice that there are significant fluctuations as the size of the lesion increases. This is because the skin lesion is not always placed directly in the center of the image, thus capturing a portion of the skin lesion throwing off the performance. An example of this can be shown in row 4 in Fig. 5 where the skin lesion is relatively small but it is placed directly where both the *Center Cropped* and *Structured Patches* are focused causing a less accurate skin capture. The results for ISIC 2017 in Fig. 9 and ISIC 2018 in Fig. 9 also align closely with the results from the ISIC 2018 dataset.
- In Figs 11, 12 and 10 evaluated the *classification* problem. In all confusion matrices, *Center Cropped* and *Structured Patches* approaches performed better for Types 1, 2, and 3, which is lighter skin. The result had more skin-focused approaches trying to avoid capturing the skin lesion. Comparatively, the *Full image* and *random patches* approach, marginal performed better on Types 5 and 6, which is the darker skin. This result happened because the *Center Cropped* and *Structured Patches* captured the skin lesions or other artifacts, thus skewing the results. For each confusion matrix there is a metrics table showing the performance for ISIC 2018 on Tables 14, 15, 16, and 17. Both the *Center Cropped* and *Structured Patches* across all the metrics are outperforming *Full image* and *random patches*. In ISIC 2016 dataset, in Tables 6, 7, 8 and 9, the baseline results only mapped 18 out of 1277 between Fitzpatrick skin types 2–6 totaling to 1.4%. The each of the four approaches performance was impacted since only missing a few would be a large percentage of the subset.

In summary, one could safely select and adopt the *Center Cropped* approach across all datasets based on the overall performance.

7 Conclusion

The four approaches proposed in this paper help solve the problem of datasets that lack Fitzpatrick skin type classification. In these approaches, we used dermoscopic images and applied different strategies for computing the ITA value as a proxy for the skin type. Our baseline was the image where the skin lesion was removed using the provided pixel masks and then the ITA was computed. All approaches were compared against the ITA value from this baseline. All ITA values were then used to map to conversion Table 2 and compared against the baseline's mapped classification.

The *Center Cropped* and *Structured Patches* approaches showed some promise from an ITA calculation in Fig. 10. As long as the skin lesion was less than 80% of the image, the same two approaches performed the best shown in Fig. 7.

Based on these findings, we propose computing the ITA values using both *Center Cropped* and *Structured Patches* to compute the ITA values and then convert them to Fitzpatrick skin type classes using Table 2. These techniques could aid in evaluating datasets to

see if there are any Fitzpatrick skin type imbalances. This metadata could also help evaluate possible fairness concerns in AI/ML models.

One of the limitations of this work is that the ISIC datasets do not provide the Fitzpatrick metadata, making our experiments not as strong as datasets that might provide this information. Comparing similar works may not be fair because many do not use the same actual baseline or ground truth. Another limitation is that this experiment was tested solely on dermoscopic images and would likely have poor performance on the Fitzpatrick17k dataset, including irrelevant backgrounds, full-body shots, or limb shots, making our approaches less effective. Last, we did not try to remove stickers or other pen marks visible on the subject's skin, which could have changed the estimate.

Future work could include having a more intelligent way of selecting the patches, such as a fitness test, to avoid skin lesions. We could also find ways to remove the stickers and pen markings to see how much effect they have on the results. Another experiment could be using Mask R-CNN to create the skin lesion mask instead of pixel mask annotation. As a final test, we could implement ML solutions to estimate the ITA values and Fitzpatrick skin type and see how well they compare to our methods.

Declarations

The authors did not receive support from any organization for the submitted work.

Competing interests All authors certify that they have no affiliations with or involvement in any organization or entity with any financial interest or non-financial interest in the subject matter or materials discussed in this manuscript.

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